

Quantiles and Ironing: A Guided Reading

Pause at each check to make a prediction, calculate an example, or complete part of the proof. Write in the space after each question. If you get stuck, use the hint on the indicated page; then compare your reasoning with the worked solution before continuing. Hints and solutions are collected at the end, and all the reasoning needed to continue is supplied here.

The problem we are trying to solve

We have one item and buyers with independent private values $V_i \sim F_i$. An allocation rule $x_i(\vec{v})$ is buyer i 's probability of receiving the item; $p_i(\vec{v})$ is their expected payment. Their utility is $v_i x_i(\vec{v}) - p_i(\vec{v})$. A mechanism is *dominant-strategy incentive compatible* (DSIC) if truthful reporting maximizes each buyer's expected utility for every fixed profile of other reports. Expectations in this definition are over the mechanism's random choices.

For this reading, assume each distribution has support $[0, v_{\max}]$, is absolutely continuous, and has positive density f on the interior, apart from possible breakpoints. We suppress the buyer's index when studying one buyer. In an auction, the one-buyer calculations hold with the other bids fixed.

We use two facts from single-parameter mechanism design. First, a DSIC allocation must be weakly increasing in the buyer's reported value. Conversely, any such allocation has normalized DSIC payments

$$p(v) = vx(v) - \int_0^v x(z) dz, \quad p(0) = 0.$$

We require *individual rationality* (IR): truthful utility is nonnegative for every fixed profile of other bids. IR forces $p(0) \leq 0$; for a fixed allocation, revenue is maximized by setting $p(0) = 0$, which preserves DSIC and gives utility $\int_0^v x(z) dz \geq 0$. Thus normalization is without loss among DSIC, IR mechanisms.

For a deterministic allocation that starts serving at value t , this charges the winner t , their *critical value*.

Second, expected revenue equals expected *virtual welfare*:

$$\mathbb{E} \left[\sum_i p_i(\vec{V}) \right] = \mathbb{E} \left[\sum_i \varphi_i(V_i) x_i(\vec{V}) \right], \quad \varphi(v) = v - \frac{1 - F(v)}{f(v)}.$$

The function φ is the buyer's *virtual value*. It is a way to account for expected revenue, not the buyer's payment at that value. These two facts are the starting point; we will derive the ironing argument from them.

Pointwise, virtual welfare is maximized by serving a buyer with the highest nonnegative virtual value. But if $\varphi(v)$ decreases somewhere, this rule can turn a win into a loss when the buyer raises their value. How can we maximize revenue while keeping the allocation monotone?

1. Index the same buyer by their quantile

An agent's *tail quantile* is the fraction of the distribution with higher values:

$$q = 1 - F(v), \quad v(q) = F^{-1}(1 - q).$$

The inverse demand curve $v(q)$ gives the corresponding value. For $U[0, 10]$, value 9 is in the top 10%, so $q = 0.1$. Smaller quantiles mean higher values: $q = 0$ is the top of the support, and $q = 1$ is the bottom.

Check 1: Which direction is “higher”?

For $V \sim U[0, 10]$:

- a. Find the quantile of value 2, and the value at quantile 0.3.

- b. Does “in the top 30%” mean $Q \leq 0.3$ or $Q \geq 0.3$?

- c. Now let $V \sim F$ and $Q = 1 - F(V)$. Express $\Pr[Q \leq a]$ as an event about V , and show that it equals a for $0 \leq a \leq 1$.

Hint: p. 12. Worked solution: p. 13.

Thus $Q \sim U[0, 1]$. The values need not be uniform; their *positions in their own distribution* are. For any integrable quantity h ,

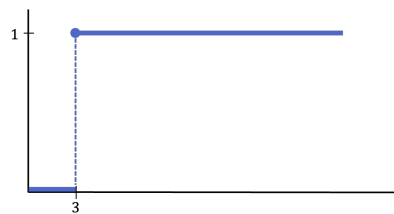
$$\mathbb{E}[h(V)] = \int_0^{v_{\max}} h(v)f(v) dv = \int_0^1 h(v(q)) dq.$$

Equal-width quantile intervals contain equal probability. The density factor has already been accounted for, while F still determines $v(q)$.

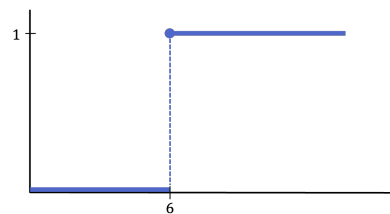
The same allocation in these coordinates is $y(q) = x(v(q))$. Because x is weakly increasing in value, y is weakly *decreasing* in quantile. Keep this reversal in mind whenever a derivative appears below.

2. An allocation can be a mixture of prices

A posted price t serves the buyer exactly when $v \geq t$. Now draw a price independently of the buyer's report: post 3 with probability $1/3$ and 6 with probability $2/3$.



Price 3



Price 6

Check 2: What does the buyer actually receive and pay?

Find the allocation probability and expected payment, conditional on each value range:

Value	Allocation probability	Expected payment
$v < 3$		
$3 \leq v < 6$		
$v \geq 6$		

The random price has mean 5. Is this the same allocation as posting a fixed price of 5? Give one value that distinguishes them.

Hint: p. 12. Worked solution: p. 13.

We are mixing prices and their corresponding sale probabilities with the *same weights*. If a price $v(a)$ sells with probability a and a price $v(b)$ sells with probability b , mixing them with probabilities α and $1 - \alpha$ gives sale probability $\alpha a + (1 - \alpha)b$.

Why every monotone allocation has this interpretation

For a weakly increasing allocation $x(v) \in [0, 1]$, choose a random price T whose CDF agrees with $x(v)$ at its continuity points. Then the probability of service at value v is exactly $x(v)$. A jump in x is a point mass on that price; where x has a continuous density of increases, the price density is $x'(v)$. Allow mass at price zero and a never-serve option when needed. At a discontinuity, choose acceptance at equality to match the specified allocation.

Every fixed posted price is DSIC. Drawing the price independently preserves DSIC, and the normalized payment identity gives the same expected payment as the original allocation. In quantile

space, the corresponding price-quantile distribution has CDF $G^y(q) = 1 - y(q)$ in the interior: service occurs when the price's quantile exceeds the buyer's quantile. Endpoint masses and acceptance at equality give the same conventions as above.

This representation is for one buyer, holding other bids fixed. It does not license drawing independent prices for all buyers and ignoring the single-item constraint.

3. Revenue becomes a curve whose slopes we understand

The expected revenue from a posted price can be indexed either by its value or by its sale probability:

$$R(v) = v[1 - F(v)], \quad P(q) = qv(q) = R(v(q)).$$

Here $P(q)$ is the revenue from the price accepted with probability q . Under our support assumptions, $P(0) = P(1) = 0$.

Check 3: Differentiate the revenue curve

- a. Differentiate $F(v(q)) = 1 - q$ to find $v'(q)$. Then differentiate $P(q) = qv(q)$ and show that

$$P'(q) = \varphi(v(q)).$$

- b. A distribution is *regular* when $\varphi(v)$ is weakly increasing in value. Does this make P concave or convex? Explain the direction using quantiles.

Hint: p. 12. Worked solution: p. 13.

Regularity is exactly concavity of P : virtual values increase as values increase, so the slopes $P'(q)$ decrease as quantiles increase. Concavity here is a statement about the curve in *quantile space*. For $U[0, 10]$, for example, $P(q) = 10q(1 - q)$ and $P'(q) = 10 - 20q = 2v(q) - 10$.

The cumulative virtual value is therefore just the revenue curve:

$$\Phi(q) = \int_0^q \varphi(v(t)) dt = P(q).$$

This is the useful representation: we can repair the *slopes* by changing the shape of the revenue curve.

4. Find the problem in the bimodal example

Return to the bimodal buyer from the notes:

$$V \sim \begin{cases} U[0, 3] & \text{with probability } 3/4, \\ U(3, 8] & \text{with probability } 1/4. \end{cases}$$

The density is $1/4$ on the first interval and $1/20$ on the second. Thus

$$1 - F(v) = \begin{cases} 1 - v/4 & 0 \leq v \leq 3, \\ (8 - v)/20 & 3 < v \leq 8, \end{cases} \quad \varphi(v) = \begin{cases} 2v - 4 & 0 < v < 3, \\ 2v - 8 & 3 < v < 8. \end{cases}$$

Check 4: Locate the failure in both coordinate systems (decimals are fine)

- a. With only this buyer, would raw virtual-welfare maximization serve them at $v = 2.5$? At $v = 3.5$? Why does this fail monotonicity?

- b. The top quarter of values comes from $(3, 8]$. Invert the two expressions for $q = 1 - F(v)$ to find $v(q)$, then multiply by q to find $P(q)$.

- c. Find the slopes of P just before and just after $q = 1/4$. What prevents concavity?

Hint: p. 12. Worked solution: p. 14.

The worked calculation gives

$$P(q) = \begin{cases} 8q - 20q^2 & 0 \leq q \leq 1/4, \\ 4q - 4q^2 & 1/4 \leq q \leq 1. \end{cases}$$

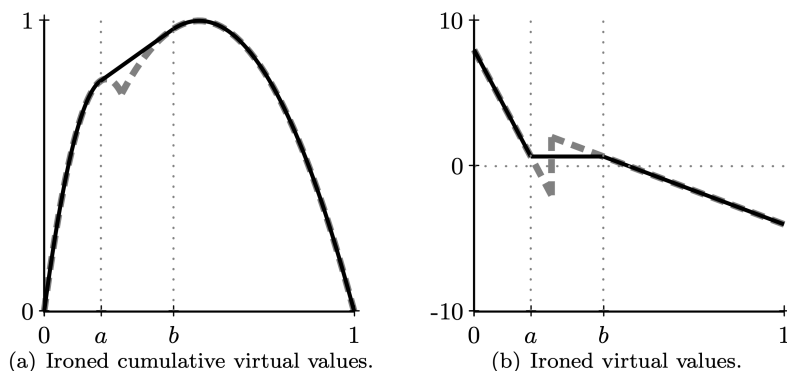
Its slope jumps upward at $q = 1/4$. Each individual piece is concave; the way the pieces meet creates the problem.

5. Iron the curve, then differentiate back

Let $\bar{P}$ be the *least concave majorant* of P : the smallest concave function lying weakly above it. On an open interval (a, b) where $\bar{P} > P$, the hull is a straight line between contact points $(a, P(a))$ and $(b, P(b))$. Define ironed virtual values by

$$\bar{\varphi}(v(q)) = \bar{P}'(q).$$

At corners, choose consistent one-sided slopes; the exact choices at these probability-zero types do not affect expected revenue. Since $\bar{P}$ is concave, $\bar{\varphi}$ is weakly increasing in value.



Both horizontal axes show quantile. In this figure from the notes, the dashed curves are the original revenue curve and virtual values, and the solid curves are ironed. The straight segment on the left becomes a flat interval on the right.

Check 5: What exactly gets averaged?

- a. Write the slope of the straight segment from a to b in terms of $P(a)$ and $P(b)$. Use $P' = \varphi \circ v$ to express this slope as an average of virtual values.

- b. Does $\bar{P} \geq P$ imply $\bar{\varphi}(v) \geq \varphi(v)$ for every value? Use the right-hand plot to decide.

Hint: p. 12. Worked solution: p. 14.

For the bimodal example, the contact quantiles and common slope are

$$a = \frac{5 + \sqrt{5}}{40}, \quad b = \frac{1 + \sqrt{5}}{8}, \quad s = 3 - \sqrt{5}.$$

The corresponding value interval is

$$[v(b), v(a)] = \left[\frac{7 - \sqrt{5}}{2}, \frac{11 - \sqrt{5}}{2} \right] \approx [2.382, 4.382].$$

Throughout this interval, $\bar{\varphi}(v) = s \approx 0.764$. Outside it, the virtual values are unchanged. The endpoint derivation in the worked solution to Check 5 can be skipped on a first pass; the next steps only require these displayed values.

6. Why ironing gives an upper bound

Fix a monotone allocation $y(q)$. We compare its expected virtual welfare with its expected *ironed* virtual welfare. The allocation stays fixed; only the objective changes. We cannot compare virtual values pointwise, as the previous check showed.

Check 6: Make the comparison using integration by parts

First suppose y is continuously differentiable. Starting from

$$\mathbb{E}[\varphi(v(Q))y(Q)] = \int_0^1 P'(q)y(q) dq,$$

a. Integrate by parts. Why do the boundary terms vanish?

b. What is the sign of $-y'(q)$? Use that sign to compare the result with the same integral containing $\bar{P}$.

Check 6 (continued)

- c. Subtract the two objectives. What condition on the allocation makes the difference zero wherever $\bar{P} > P$?

Hint: p. 12. Worked solution: p. 15.

The resulting theorem is

$$\mathbb{E}[\varphi(v(Q))y(Q)] \leq \mathbb{E}[\bar{\varphi}(v(Q))y(Q)],$$

with equality whenever y is constant on each open ironed interval. In value space, the same statement is

$$\mathbb{E}[\varphi(V)x(V)] \leq \mathbb{E}[\bar{\varphi}(V)x(V)],$$

with equality when x is constant on the corresponding value intervals.

Step allocations are included. A posted price gives a jump in y , so its ordinary derivative does not record the whole allocation. A downward jump of size c at quantile t puts probability c on that posted price. It contributes $cP(t)$ to expected revenue and $c\bar{P}(t)$ to the ironed objective. More generally, the price-mixture representation proves the same inequality by averaging $P(t) \leq \bar{P}(t)$ over the random price quantile. The worked solution to Check 6 gives this argument explicitly. Equality allows jumps at contact endpoints, where the curves agree.

7. Attain a point on the ironed curve

A point of $\bar{P}$ above P corresponds to a mixture of the two contact prices. As before, mixing these prices and mixing their sale probabilities are the same mechanism, with the same weights.

Check 7: Choose the mixture to hit a sale probability

Fix $q \in (a, b)$. Post $v(a)$ with probability α and $v(b)$ with probability $1 - \alpha$.

- a. Solve for α so the probability of sale is q .
- b. Calculate the expected revenue. Why is it $\bar{P}(q)$?
- c. For the bimodal example at $q = 1/4$, compare this mixture with the fixed price $v(1/4) = 3$, which earns $3/4$. Use the endpoints given above. Decimals are fine.

Hint: p. 12. Worked solution: p. 15.

The mixture is DSIC because the random price does not depend on the buyer's report. Its allocation is constant between the two prices: these types buy only when the lower price is drawn. That is exactly the equality condition in the ironing bound.

This construction holds the *sale probability* fixed. Without a constraint on the sale probability, a single buyer's mixture cannot beat their best fixed price: its revenue is an average of fixed-price revenues. In the auction below, the critical price is fixed once the other bids are fixed, and random before they are drawn. The allocation is flat on ironed intervals, both for fixed other bids and after averaging over them, just as in the mixture above.

8. Return to the auction

The final mechanism computes every buyer's ironed virtual value, allocates the item to a buyer with the highest nonnegative score, and uses a fixed priority order to break ties. If all scores are negative, it makes no sale. Charge a winning buyer their critical value, holding other bids fixed; charge everyone else zero. The payment is a threshold in *value space*, not their ironed virtual value.

Check 8: Why does this attain the upper bound?

- a. Explain why increasing a buyer's value cannot turn a win into a loss.

- b. Fix the other bids. Why is this buyer's allocation constant on an ironed interval? What could fail if an ironed-score tie were broken by the higher reported value?

Check 8 (continued)

- c. For any feasible normalized DSIC allocation x and the proposed allocation x^* , justify each step:

$$\begin{aligned}\text{Rev}(x) &= \mathbb{E} \left[\sum_i \varphi_i(V_i) x_i(\vec{V}) \right] \\ &\leq \mathbb{E} \left[\sum_i \bar{\varphi}_i(V_i) x_i(\vec{V}) \right] \\ &\leq \mathbb{E} \left[\sum_i \bar{\varphi}_i(V_i) x_i^*(\vec{V}) \right] = \text{Rev}(x^*).\end{aligned}$$

Hint: p. 12. Worked solution: p. 16.

Final check. Buyers 1 and 2 have the bimodal distribution and realized values 2.5 and 3.5, respectively. Buyer 1 has priority in a tie. Who wins, and what do they pay? Does raising buyer 1's bid to 4 change either?

Worked solution: p. 16.

Hints

Check 1. For a uniform value on $[0, 10]$, $F(v) = v/10$. For the general part, remember that subtracting from 1 reverses the inequality, and that F is increasing.

Check: p. 2. Worked solution: p. 13.

Check 2. For each value range, ask which of the two possible prices the buyer can afford. Average the payment over the price draw, counting zero when no purchase occurs.

Check: p. 3. Worked solution: p. 13.

Check 3. The chain rule gives $f(v(q))v'(q) = -1$. A concave function has weakly decreasing slopes in its own horizontal coordinate.

Check: p. 4. Worked solution: p. 13.

Check 4. For a single buyer, raw virtual-welfare maximization serves when the virtual value is nonnegative. The first quantile branch comes from the *upper* value interval.

Check: p. 5. Worked solution: p. 14.

Check 5. The slope of a chord is its change in height divided by its change in quantile. A nonnegative gap between two functions does not order their derivatives.

Check: p. 6. Worked solution: p. 14.

Check 6. Use $u = y$ and $dw = P'(q) dq$. Both P and $\bar{P}$ vanish at 0 and 1. For equality, look at the product of the allocation's decrease and the gap between the curves.

Check: p. 7. Worked solution: p. 15.

Check 7. The high price $v(a)$ sells with probability a , and the low price $v(b)$ sells with probability b . Average these probabilities first, then average their revenues using the same weights.

Check: p. 9. Worked solution: p. 15.

Check 8. Use a separate reason for each step: the revenue identity, the ironing inequality, pointwise maximization, and the equality condition. A fixed score and a fixed priority cannot change the winner when the other bids stay fixed.

Check: p. 10. Worked solution: p. 16.

Worked Solutions

Check 1: Quantiles reverse the direction of values

For value 2, $q = 1 - 2/10 = 0.8$. At quantile 0.3, $v(0.3) = 10(1 - 0.3) = 7$. The top 30% means $Q \leq 0.3$. In general,

$$\begin{aligned}\Pr[Q \leq a] &= \Pr[1 - F(V) \leq a] \\ &= \Pr[V \geq F^{-1}(1 - a)] \\ &= 1 - F(F^{-1}(1 - a)) = a.\end{aligned}$$

Continuity means that equality at the threshold has probability zero. Thus Q has the uniform CDF on $[0, 1]$. The argument does not require V itself to be uniform.

Return to Check 1, p. 2.

Check 2: Allocation and payment under the two-price mixture

The completed table is

Value	Allocation probability	Expected payment
$v < 3$	0	0
$3 \leq v < 6$	1/3	$(1/3)3 = 1$
$v \geq 6$	1	$(1/3)3 + (2/3)6 = 5$

The buyer pays the realized posted price only if they buy. In the middle range, the expected payment is 1, but the payment conditional on receiving the item is 3. A value of 4 distinguishes the mixture from posting 5: the mixture serves with probability 1/3, while a fixed price of 5 does not serve that type. The mean of the random price does not determine its allocation rule.

Return to Check 2, p. 3.

Check 3: The derivative is a virtual value

Differentiating $F(v(q)) = 1 - q$ gives

$$v'(q) = -\frac{1}{f(v(q))}.$$

Then

$$\begin{aligned}P'(q) &= v(q) + qv'(q) \\ &= v(q) - \frac{q}{f(v(q))} \\ &= v(q) - \frac{1 - F(v(q))}{f(v(q))} = \varphi(v(q)).\end{aligned}$$

As q increases, $v(q)$ decreases. If φ is weakly increasing in value, then $P'(q)$ is weakly decreasing in quantile, so P is concave. The argument also works in reverse. At density breakpoints, compare the one-sided slopes.

Return to Check 3, p. 4.

Check 4: The upward jump in slope is the obstruction

The virtual values at 2.5 and 3.5 are 1 and -1 . The raw rule serves at the lower value and refuses at the higher value. A DSIC allocation cannot do that.

For $q \leq 1/4$, use $q = (8-v)/20$, so $v(q) = 8 - 20q$. For $q \geq 1/4$, use $q = 1 - v/4$, so $v(q) = 4 - 4q$. Therefore

$$P(q) = \begin{cases} 8q - 20q^2 & q \leq 1/4, \\ 4q - 4q^2 & q \geq 1/4. \end{cases}$$

The left derivative at $1/4$ is $8 - 40(1/4) = -2$, while the right derivative is $4 - 8(1/4) = 2$. Slopes of a concave function cannot jump upward. This is the quantile-space version of the downward jump in virtual values as the buyer's value crosses 3.

Return to Check 4, p. 5.

Check 5: Average the virtual values over quantiles

On (a, b) the ironed slope is

$$s = \frac{P(b) - P(a)}{b - a} = \frac{1}{b - a} \int_a^b \varphi(v(t)) dt.$$

Thus we replace virtual values by their average over this quantile interval. Some go up and some go down. For example, at value 2.5 the original virtual value is 1, above the ironed value $3 - \sqrt{5} \approx 0.764$. At value 3.5 the original is -1 , below the same ironed value. So $\bar{P} \geq P$ gives no pointwise ordering of the virtual values.

Find the contact endpoints (skip on a first pass). Each piece of P is a concave quadratic. A straight segment that smoothly joins them must have their common tangent slope:

$$8 - 40a = s = 4 - 8b.$$

The tangent to $8q - 20q^2$ at a has intercept $20a^2$; the tangent to $4q - 4q^2$ at b has intercept $4b^2$. The tangent lines coincide when $20a^2 = 4b^2$, so $b = \sqrt{5}a$. Solving gives

$$a = \frac{5 + \sqrt{5}}{40}, \quad b = \frac{1 + \sqrt{5}}{8}, \quad s = 3 - \sqrt{5}.$$

A concave quadratic lies below each of its tangent lines, so this chord lies above both pieces between a and b . Keeping P outside the interval now gives a concave curve. Any concave majorant of P must lie above the chord between these contact points, so this is the least one.

Return to Check 5, p. 6.

Check 6: The allocation's decreases give nonnegative weights

Integration by parts yields

$$\begin{aligned}
\int_0^1 P'(q)y(q) dq &= [P(q)y(q)]_0^1 - \int_0^1 P(q)y'(q) dq \\
&= \int_0^1 (-y'(q))P(q) dq \\
&\leq \int_0^1 (-y'(q))\bar{P}(q) dq \\
&= \int_0^1 \bar{P}'(q)y(q) dq.
\end{aligned}$$

The boundary terms vanish because $P(0) = P(1) = \bar{P}(0) = \bar{P}(1) = 0$. The inequality uses both $-y'(q) \geq 0$ and $\bar{P}(q) \geq P(q)$. Reversing the integration by parts gives the final equality. In particular,

$$\mathbb{E}[\bar{\varphi}(v(Q))y(Q)] - \mathbb{E}[\varphi(v(Q))y(Q)] = \int_0^1 (-y'(q))(\bar{P}(q) - P(q)) dq.$$

If y is constant on each open ironed interval, then wherever the gap is positive, $y' = 0$. Elsewhere the gap is zero. The difference vanishes.

The general monotone case. Let T be the random *price quantile* representing y . For a fixed price quantile t , the allocation serves $q < t$, and

$$\int_0^t P'(q) dq = P(t), \quad \int_0^t \bar{P}'(q) dq = \bar{P}(t).$$

Average over T . The two expected virtual welfares become $\mathbb{E}[P(T)]$ and $\mathbb{E}[\bar{P}(T)]$, respectively, proving the inequality without assuming a smooth allocation. If the allocation is constant on each open ironed interval, the price distribution assigns no mass there. All its mass is on contact points, so the two averages agree. Endpoint options have zero contribution under our support assumptions.

Return to Check 6, p. 7.

Check 7: The same weights mix prices, sale probabilities, and revenues

The sale probability is $\alpha a + (1 - \alpha)b$. Setting this equal to q gives

$$\alpha = \frac{b - q}{b - a}, \quad 1 - \alpha = \frac{q - a}{b - a}.$$

The expected revenue is consequently

$$\frac{b - q}{b - a}P(a) + \frac{q - a}{b - a}P(b) = \bar{P}(q),$$

the height of the chord at q .

At $q = 1/4$ in the bimodal example,

$$\alpha = \frac{5 - \sqrt{5}}{4} \approx 0.691.$$

Post the high price $v(a) = (11 - \sqrt{5})/2 \approx 4.382$ with this probability, and the low price $v(b) = (7 - \sqrt{5})/2 \approx 2.382$ otherwise. The probability of sale is still $1/4$, but the revenue is

$$\bar{P}(1/4) = \frac{9 - \sqrt{5}}{8} \approx 0.8455 > 0.75 = P(1/4).$$

The gain is possible because we hold the overall probability of sale fixed while changing which types are served and what they pay.

Return to Check 7, p. 9.

Check 8: Monotonicity makes the rule truthful; flatness makes the bound tight

Fix the other bids. Increasing buyer i 's value weakly increases their ironed score. With fixed priority, a winning buyer cannot lose by raising that score, so the allocation is monotone. The payment identity therefore implements it as a normalized DSIC mechanism.

Inside an ironed interval, the score does not change. Neither the competing scores nor the priority order changes, so the allocation stays constant. This supplies the equality condition, bidder by bidder.

A tie broken by the higher *reported value* can change the winner within the interval. For example, with the other bimodal buyer fixed at 3.5, increasing a bid from 2.5 to 3.7 changes that tie's winner even though both scores stay at $3 - \sqrt{5}$. Such a tie rule can still be monotone, but it fails the flatness condition needed to make the ironing bound tight.

The first equality in the check is the expected-revenue identity. For the first inequality, apply the ironing theorem to each buyer while fixing the other bids, then average and sum. Independence ensures that buyer i still has distribution F_i under this conditioning. The second inequality holds because x^* maximizes ironed virtual welfare for every realized profile, including the no-sale option. The last equality uses the just-established flatness and then the revenue identity. Thus x^* is revenue-optimal among feasible DSIC, IR mechanisms, since normalization loses no revenue.

Final check. Buyer 1 wins the score tie and pays $(7 - \sqrt{5})/2 \approx 2.382$, the smallest value at which they reach buyer 2's ironed score. Raising their bid to 4 keeps them in the ironed interval, so neither the winner nor the payment changes.

Return to Check 8, p. 10.

Acknowledgements

This guided reading is based on the quantiles and ironing notes from DS 574. Those notes were developed in part using materials by Tim Roughgarden and Jason Hartline, and in particular, the books “Twenty Lectures on Algorithmic Game Theory” and “Mechanism Design and Approximation” [1, 2].

References

- [1] Jason D. Hartline. Mechanism design and approximation. *Book draft. October*, 122, 2013.
- [2] Tim Roughgarden. *Twenty Lectures on Algorithmic Game Theory*. Cambridge University Press, 2016.